October University for Modern Science and Arts

Faculty of Computer Science

Machine Learning Project Proposal

Course Name: Machine Learning
Course Code: CS363
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Submission Date: March 22, 2026

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1 Project Title

Forecastify

2 Project Idea and Motivation

Sometimes retailers face challenges in setting the right prices, especially during the time when customers have more money to spend, like when SNAP (Supplemental Nutrition Assistance Program) benefits are distributed. During these times, the purchasing behavior changes, and demand for products can increase significantly.

This problem is critically important because during the high-demand periods poor pricing can either reduce the revenue or lower sales for the seller. Also, setting prices too high may drive customers away, and too low prices will mean missed opportunities for profit. So, understanding these patterns will help sellers make smarter pricing choices.

Solving this problem benefits retailers by helping them increase revenue and maintain sales volume, customers also benefit from more predictable pricing and stores can manage their inventory more efficiently.

Machine Learning is suitable because it can analyze large and complex datasets; it can detect patterns and predict demand in ways traditional methods cannot.

3 Problem Statement and Objectives

Problem Statement

Price Optimization during High-Liquidity Cycles (SNAP)

Retail sales do not depend only on prices; they are also strongly affected by external factors, such as when SNAP (Supplemental Nutrition Assistance Program) benefits are distributed among the markets. During these periods, a large number of customers have more purchasing power that ends up leading to noticeable change in buying behavior.

However, retailers still do not understand how these cycles affect the pricing strategies. Therefore, this project aims to explore that relationship by building a forecasting model that measures the impact of SNAP distribution on sales. It will also help to determine the best pricing strategy for different product categories, allowing retailers to increase revenue without reducing sales during busy periods.

- The dataset used: M5 Forecasting – Accuracy (Walmart Sales Data)

- The target variable is Daily Sales
- Type of task:
  - This is a Regression problem as it's to predict a numerical value

Measurable Objectives

- Achieve high regression accuracy using RMSE and R^2 metrics.
- Compare at least five ML models (XGBoost, LightGBM, Prophet, Random Forest, TFT).
- Identify the most influential features affecting sales during SNAP periods.
- Build a forecasting model that measures the impact of SNAP distribution on sales.
- Determine the best pricing strategy for different product categories.

4 Dataset

Dataset Source

- Dataset name: M5 Forecasting – Accuracy (Walmart Sales Data)
- Source: Kaggle
- Link: `https://www.kaggle.com/c/m5-forecasting-accuracy`

Dataset Characteristics

- Number of samples: The full dataset contains over 50 million rows
- Number of features: Approximately 20–25 features (including lags and event impacts)
- Feature types:
  - Numerical: `sales`, `sell_price`, `lag_1`, `lag_7`, `event_impact`
  - Categorical: `store_id`, `item_category`, `item_sub_category`, `year`, `month`, `wday`
- Target variable: `sales` (the number of units sold for a specific product on a specific day)

Preprocessing Summary

- Missing value handling: We used Data Trimming to remove "leading zeros" (days before a product was first stocked). Empty values created by lag features were either filled with 0 or the rows were removed to keep the data clean.
- Encoding techniques: Categorical data like `store_id` and `item_category` are typically Label Encoded or One-Hot Encoded for the model.
- Normalization/standardization: We scaled the `sell_price` and `event_impact` features. This ensures that a price of $20.00 doesn't overpower a sales count of 1 in the model's mathematical calculations.
- Feature selection: We selected key "Time-Series" features. Instead of just using a raw date, we used `wday` (day of week), `month`, and `snap` (food stamp days). We also added Lag Features (`lag_1`, `lag_7`) to tell the model what happened yesterday and last week.

Dataset Samples

Show Samples of the Dataset:

- Data Distribution graph
- Images Samples *if it's images problem*
- Statistics graphs *if Numeric data*

Dataset Limitations

- Class imbalance: refers to the fact that 0 and 1 sales happen much more often than any other number. The Problem: Because the data is so heavily biased, the model might become biased. It may learn to predict low numbers perfectly but fail to predict the big numbers (like 10 or 20 units) because it hasn't seen enough examples of them.
- Noisy or missing data: "0" in the data could mean nobody wanted the item, or it could mean the item was out of stock (not exist). Since we don't have "Out-of-Stock" labels, the model might incorrectly think people stopped liking a product when it was actually just missing from the store.
- Bias in data collection:

- Geographic Bias: The shopping patterns in California are different from those in Texas. A model trained on this data might not work for a store in London or New York.
- Time Bias: The data covers a specific period. If there was a major economic change or a global event (like a pandemic) during those years, the "normal" patterns would be distorted.
- Event Bias: We only have "SNAP" data for these specific states; other regions use different systems that aren't captured here.

5 Literature Review

Paper 1: Sales Forecasting in Fashion Retailing – A Review

- **Problem addressed:** Traditional statistical methods struggle to capture non-linear patterns and handle the high number of variables (size, color, price) typical in women's apparel sales.
- **Models used:** The study compared Seasonal Single Exponential Smoothing (SSES), Winters' three-parameter model, and an Artificial Neural Network (ANN).
- **Performance achieved:** The ANN showed a good coefficient of R^2, but a low correlation coefficient (R) between actual and forecasted sales. Neuro-Fuzzy Models achieved a 47.5 MdAPE (Median Absolute Percentage Error), significantly lower than the 52.3 of traditional HWS. Extreme Learning Machine (ELM) with adaptive metrics (AD-ELM) reduced error to a 6.02% MAPE, compared to 15.33% for standard ANN.
- **Limitations:** The low correlation was attributed to "noisy" data. Additionally, early NN models faced challenges due to limited past sales data.
- **Relation to our project:** Reliable predictions prevent "customer store switching," a long-term problem where shoppers move to competitors because their desired items are unavailable.

[1]

Paper 2: Forecasting German Car Sales Using Google Data and Multivariate Models

Guyon, H., & Petiot, J.-F. (2011). Market share predictions: a new model with rating-based conjoint analysis. *International Journal of Market Research*, 53(6), 831–857.

- **Problem addressed:** The automotive industry's critical need for accurate long-term forecasting (up to 24 months), which is vital due to lengthy 12-to-60-month production cycles. While most existing research focuses on short-term horizons, this study investigates whether incorporating Google Trends data as a leading indicator can reduce financial risks like overstocking or shortages by improving the predictive accuracy of long-term sales models.
- **Models used:** Vector Auto-Regressive (VAR/VARX) & Error Correction Models: these multivariate models analyze how multiple variables (like car sales and Google search trends) influence each other over time. They are designed to capture both immediate fluctuations and stable, long-term relationships between economic data and consumer behavior. Bayesian and Nonlinear Models: These advanced techniques, such as BVAR and LSTAR, allow the researchers to handle complex data without overcomplicating the math.
- **Performance achieved:** Long-Term Superiority of Digital Data: While simple traditional models are effective for short-term predictions (under 10 months), models incorporating Google Trends become significantly more accurate for long-term horizons (up to 24 months).
- **Limitations:** Sampling Variability: Google data is based on a sample rather than the full population of searches. This variability can lead to inconsistent results in high-dimensional VEC models.
- **Relation to our project:** Both the paper and our dataset aim to solve the problem of predicting future sales. While the paper focuses on the German automotive industry, our M5 dataset (originally from Walmart) focuses on retail product sales. The underlying goal—reducing uncertainty in the supply chain to prevent overstocking or shortages—is identical.

[2]

Paper 3: Range Unit-Root (RUR) Tests

Aparicio, F., A. Escribano, and A. Garcia (2006): "Range Unit-Root (RUR) tests: robust against nonlinearities, error distributions, structural breaks and outliers," *Journal of Time Series Analysis*, 27(4), 545–576.

Paper Analysis: It highlights a critical trade-off for data-driven projects: while economic variables are theoretically important, parsimonious models using high-frequency digital data often yield higher out-of-sample accuracy for long horizons because they avoid the "curse of dimensionality" and over-parameterization.

[3]

6 Proposed Models

1. XGBoost (Extreme Gradient Boosting)

- **Mathematical Intuition:** XGBoost is an implementation of gradient-boosted decision trees designed for speed and performance. It minimizes a regularized objective function:

$$\text{Obj}(\theta) = L(\theta) + \Omega(\theta) \tag{1}$$

- **Key Hyperparameters:**
  - `learning_rate`: Scales the contribution of each tree to prevent overfitting.
  - `max_depth`: Controls the depth of trees to capture complex feature interactions (e.g., price $\times$ snap).
  - `lambda` (L_2 regularization) / `alpha` (L_1 regularization): Encourages weights to be small, smoothing the prediction.
- **Suitability:** XGBoost is highly effective for tabular retail data because it inherently captures non-linear relationships and interactions between categorical variables (like `store_id`) and continuous variables (like `sell_price`) without requiring extensive feature scaling.

2. LightGBM (Light Gradient Boosting Machine)

- **Mathematical Intuition:** LightGBM optimizes the boosting process through Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). Unlike traditional level-wise growth, LightGBM uses a leaf-wise (best-first) growth strategy. It chooses the leaf with the maximum loss reduction to grow, which can significantly reduce training loss compared to level-wise algorithms.
- **Key Hyperparameters:**
  - `num_leaves`: The primary factor in controlling model complexity; it must be tuned in relation to `max_depth`.
  - `feature_fraction`: Randomly selects a subset of features for each iteration, reducing the influence of noise in high-dimensional retail datasets.

- `cat_smooth`: Helps the model handle categorical features by reducing the impact of categories with low observation counts.

- **Suitability:** Given the high cardinality of retail datasets (thousands of unique item-store combinations), LightGBM's efficiency in memory usage and training speed makes it the industry standard for large-scale forecasting competitions.

3. **Prophet (Additive Regression Model)**

- **Mathematical Intuition:** Prophet frames time-series forecasting as a curve-fitting exercise rather than a stochastic process. It decomposes the signal into four components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \tag{2}$$

- **Key Hyperparameters:**
  - `changepoint_prior_scale`: Determines the flexibility of the trend; higher values allow the model to fit more rapid changes in sales trajectory.
  - `holidays_prior_scale`: Controls the "strength" of the holiday component (crucial for `event_impact` features).
  - `seasonality_mode`: Can be 'additive' or 'multiplicative' (retail sales often show multiplicative patterns where SNAP effects scale with baseline volume).

- **Suitability:** Prophet is uniquely suited for this problem because it allows for the explicit modeling of "Holiday Effects." By feeding the `is_event` and `snap` columns into the holiday component, the model can isolate the specific uplift generated by these dates.

4. **Random Forest Regressor**

- **Mathematical Intuition:** Random Forest utilizes Bootstrap Aggregating (Bagging) to build an ensemble of independent decision trees. Each tree is trained on a random sample of the data and a random subset of features. The final prediction is the average of all individual tree outputs:

$$\hat{f} = \frac{1}{B}\sum_{b=1}^{B} f_b(x) \tag{3}$$

- **Key Hyperparameters:**
  - `n_estimators`: The number of trees; generally, more trees improve stability until a point of diminishing returns.
  - `max_features`: The size of the random subsets of features to consider when splitting a node, ensuring trees remain decorrelated.
- **Suitability:** It serves as a robust baseline. Because it does not rely on gradient descent, it is less sensitive to hyperparameter tuning and provides a reliable measure of Feature Importance, helping to validate which factors (Price vs. SNAP) drive the most variance.

5. Temporal Fusion Transformer (TFT)

- **Mathematical Intuition:** The TFT is a deep learning architecture that combines LSTM layers for local processing with Multi-Head Attention for long-term dependencies. It uses Variable Selection Networks to ignore irrelevant features and Gated Residual Networks (GRN) to skip unused components of the architecture, providing a highly adaptive learning process.
- **Key Hyperparameters:**
  - `attention_head_size`: Controls the number of different "temporal patterns" the model can track simultaneously.
  - `dropout_rate`: Essential for preventing overfitting in deep neural networks.
  - `learning_rate`: Often paired with a "scheduler" to reduce the rate as the model converges.
- **Suitability:** TFT is state-of-the-art for multi-horizon forecasting. It is specifically designed to handle "static covariates" (like `item_category`) alongside "time-varying observables" (like `sales`) and "known future inputs" (like upcoming `snap` days and `is_weekend`), making it the most sophisticated choice for this project.

7 Evaluation and Comparison Strategy

For Regression:

- MSE
- RMSE

- MAE
- R^2

These metrics are appropriate because they provide a complete picture of model performance. MAE gives us a simple average of our errors, while MSE and RMSE ensure that we are penalized for large misses in high-volume periods. Finally, R^2 allows us to measure how much of the complex sales patterns—such as seasonality and events—our model has successfully learned.

8 Teamwork and Task Delegation

The following table must be included exactly as required:

Student Name	**Student ID**	**Assigned Role**	**Main Responsibilities**	**Deliverables Contributed**
Muhammad Kandil	245629	Team Lead / Coordinator	Overall coordination, idea consolidation	Final idea & structure
Retag Ahmed	248727	Literature Review Lead	Search & summarize related work	Related work section
Sandra Philip	243369	Dataset Analyst	Dataset search, analysis & description	Dataset section
Farah Kinaz	248311	Problem Definition Lead	Problem formulation & objectives	Problem statement
All members	xxxxxx	Editor / Quality Control	Formatting, references, coherence	Final editing

References

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